📘 AI & Data Science – Assignment 6

Total Marks: 20

Question 1 (2 Marks)

Open **ChatGPT** and paste the **project prompt you created in Assignment 5**.

Communicate the **full vision of your project** to ChatGPT, including:

• The complete purpose of the application
• All main features
• The pages of the website
• The type of users (admin, normal user, etc.)
• The type of data that needs to be stored

ChatGPT should clearly understand the **entire structure of your application and database needs**.

Question 2 (5 Marks)

Ask ChatGPT to generate **complete SQL code for your database** for use with **Supabase**.

The SQL code must include:

• All database tables
• Relationships between tables
• Primary keys and foreign keys
• User roles (if needed)
• Row Level Security (RLS) policies
• Insert permissions
• Select permissions

 Important:
The SQL must be **one complete script that can be copied and pasted once**.

Save this SQL code in your GitHub repository inside a folder named:

```
props
```

Create a file called:

```
database.md
```

Paste the full SQL code inside this file.

Question 3 (3 Marks)

Go to **Supabase** and open the **SQL Editor**.

Run the SQL code you generated in ChatGPT.

This should automatically create:

• All tables
• Relationships
• Security policies
• Database structure for your project

Question 4 (2 Marks)

Inside **Supabase**, locate the following project credentials:

• **Project URL**
• **Public API Key (anon public key)**

These will be used to connect your frontend to the database.

Question 5 (3 Marks)

Open your project again in **Lovable**.

Ask Lovable to:

• Create the **Supabase configuration file**
• Connect the frontend to your Supabase project
• Use the **Project URL** and **Public API Key**

Lovable should generate the correct configuration file automatically.

Question 6 (3 Marks)

Copy the **same SQL code** from your `database.md` file and paste it into **Lovable**.

Ask Lovable to:

• Modify the frontend
• Connect pages to the database

• Fetch data from Supabase
• Display the data dynamically on the website

Your site should now **load real data from the database**.

Question 7 (2 Marks)

Return to **ChatGPT** and ask it to generate **SQL INSERT statements** to populate your database with **sample data**.

Examples of sample data may include:

• Users
• Products
• Orders
• Comments
• Posts
• Shoes (for ecommerce projects)

Run this SQL in **Supabase SQL Editor** to fill your tables with test data.

Important Instructions

• By **next Saturday**, your project must have:

- A **fully working frontend**
- A **connected Supabase database**
- **Real data loading from the database**

• Use **the Online classes** to collaborate and troubleshoot.

• On **Saturday**, every student must come with a **fully functional live project connected to Supabase**.